Covariant and Contravariant Bases, Metrics and General Curvilinear Calculus

Part 4 of the OT Science Formula Completion Workbook

OT Science Notes

2026-06-24

Contents

1 Covariant and Contravariant Bases, Metrics and General Curvilinear Calculus

This part connects coordinate geometry with tensor notation. The key idea is that a vector itself is geometric, but its components depend on the basis used to describe it. Covariant and contravariant components are two related ways of recording the same geometric object.

1.1 Coordinate Basis and Reciprocal Basis

Let

$$\mathbf{r} = \mathbf{r}(q^1, q^2, q^3).$$

The covariant basis vectors are tangent to coordinate curves:

$$\mathbf{e}_i = \frac{\partial \mathbf{r}}{\partial q^i}.$$

The contravariant basis vectors form the reciprocal basis:

$$\mathbf{e}^i \cdot \mathbf{e}_j = \delta^i_j.$$

Covariant and Contravariant Bases

$$\mathbf{e}_i = \frac{\partial \mathbf{r}}{\partial q^i}, \qquad \mathbf{e}^i = \nabla q^i.$$

$$\mathbf{e}^i \cdot \mathbf{e}_j = \delta^i_j.$$

$$\mathbf{A} = A^i \mathbf{e}_i = A_i \mathbf{e}^i.$$

$$A^i = \mathbf{A} \cdot \mathbf{e}^i, \qquad A_i = \mathbf{A} \cdot \mathbf{e}_i.$$

1.2 Metric Tensor

The metric tensor records dot products between basis vectors. It is the object that turns coordinate changes into physical lengths and angles.

Metric Tensor and Raising/Lowering

$$g_{ij} = \mathbf{e}_i \cdot \mathbf{e}_j, \qquad g^{ij} = \mathbf{e}^i \cdot \mathbf{e}^j.$$

$$g^{ik} g_{kj} = \delta^i_j.$$

$$A_i = g_{ij} A^j, \qquad A^i = g^{ij} A_j.$$

$$ds^2 = g_{ij}\, dq^i dq^j.$$

$$\mathbf{A} \cdot \mathbf{B} = g_{ij} A^i B^j = g^{ij} A_i B_j = A_i B^i = A^i B_i.$$

1.3 Orthogonal Coordinates as a Special Case

If the coordinate system is orthogonal, the metric tensor is diagonal. This is why scale factors are enough to describe many engineering coordinate systems.

Orthogonal Metric and Scale Factors

$$g_{ij} = h_i^2 \delta_{ij} \quad \text{no sum on } i.$$

$$g^{ij} = \frac{1}{h_i^2} \delta^{ij} \quad \text{no sum on } i.$$

$$\mathbf{e}_i = h_i \hat{\mathbf{e}}_i, \qquad \mathbf{e}^i = \frac{1}{h_i} \hat{\mathbf{e}}_i.$$

$$A_i = h_i A_{\text{phys},i}, \qquad A^i = \frac{A_{\text{phys},i}}{h_i}.$$

1.4 Jacobian, Metric and Volume

For a coordinate transformation $x^a = x^a(q^1, q^2, q^3)$, the Jacobian matrix connects small coordinate changes to small physical displacements.

Jacobian and Volume Element

$$J^a{}_i = \frac{\partial x^a}{\partial q^i}.$$

$$g_{ij} = \frac{\partial x^a}{\partial q^i} \frac{\partial x^a}{\partial q^j} = J^a{}_i J^a{}_j.$$

$$g = \det(g_{ij}), \qquad \sqrt{g} = |\det(J)|\,.$$

$$dV = \sqrt{g}\, dq^1 dq^2 dq^3.$$

For orthogonal coordinates,

$$\sqrt{g} = h_1 h_2 h_3.$$

1.5 Gradient, Divergence and Laplacian in General Curvilinear Coordinates

The following formulas use the metric determinant $g = \det(g_{ij})$. They are especially useful when coordinates are not necessarily orthogonal.

General Curvilinear Formulae

$$\nabla f = (\partial_i f) \mathbf{e}^i.$$

$$\nabla \cdot \mathbf{A} = \frac{1}{\sqrt{g}} \frac{\partial}{\partial q^i} \left(\sqrt{g}\, A^i \right).$$

$$\nabla^2 f = \frac{1}{\sqrt{g}} \frac{\partial}{\partial q^i} \left(\sqrt{g}\, g^{ij} \frac{\partial f}{\partial q^j} \right).$$

For orthogonal coordinates,

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \sum_{i=1}^{3} \frac{\partial}{\partial q^i} \left(\frac{h_1 h_2 h_3}{h_i^2} \frac{\partial f}{\partial q^i} \right).$$

1.6 Cylindrical and Spherical Laplacians

Common Laplacians

Cartesian:

$$\nabla^2 f = f_{xx} + f_{yy} + f_{zz}.$$

Cylindrical:

$$\nabla^2 f = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial f}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}.$$

Spherical:

$$\nabla^2 f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2\theta} \frac{\partial^2 f}{\partial \phi^2}.$$

Quick Drills and Practice: Covariant/Contravariant Components and Metrics

Level A: Conceptual recall

1. Complete: $\mathbf{e}_i =$ ____________________.
2. Complete: $\mathbf{e}^i =$ ________________.
3. Complete: $\mathbf{e}^i \cdot \mathbf{e}_j =$ __________.
4. Complete: $g_{ij} =$ ____________________.
5. Complete: $g^{ik} g_{kj} =$ __________.
6. Complete: $A_i =$ ________________.
7. Complete: $A^i =$ ________________.
8. Complete: $ds^2 =$ ____________________.

Level B: Metric computations

1. Compute g_{ij} for cylindrical coordinates directly from $\mathbf{r}(\rho, \phi, z)$.
2. Compute g^{ij} for cylindrical coordinates.
3. Compute g_{ij} for spherical coordinates directly from $\mathbf{r}(r, \theta, \phi)$.
4. Compute g^{ij} for spherical coordinates.
5. Compute $\sqrt{g}$ for Cartesian, cylindrical and spherical coordinates.
6. Show that $dV = \sqrt{g}\, dq^1 dq^2 dq^3$ gives the correct cylindrical and spherical volume elements.

Level C: Components

1. In cylindrical coordinates, if the physical components are $(A_\rho, A_\phi, A_z) = (2, 3, 4)$, write the contravariant components A^i.
2. In spherical coordinates, if the physical components are $(A_r, A_\theta, A_\phi) = (1, 2, 3)$, write A^i and A_i.
3. Given $g_{ij} = \mathrm{diag}(1, r^2, r^2 \sin^2\theta)$, lower the index of $A^i = (r, \sin\theta, 1)$.
4. Given $g^{ij} = \mathrm{diag}(1, 1/r^2, 1/(r^2 \sin^2\theta))$, raise the index of $A_i = (r^2, r^2\theta, r^2 \sin\theta)$.

Level D: Differential operators

1. Derive the cylindrical divergence formula from $\nabla \cdot \mathbf{A} = \frac{1}{\sqrt{g}}\partial_i(\sqrt{g}A^i)$.
2. Derive the spherical divergence formula from $\nabla \cdot \mathbf{A} = \frac{1}{\sqrt{g}}\partial_i(\sqrt{g}A^i)$.
3. Derive the cylindrical scalar Laplacian from the general curvilinear formula.
4. Derive the spherical scalar Laplacian from the general curvilinear formula.
5. Compute $\nabla^2 f$ in cylindrical coordinates for $f = \rho^2 + z^2$.
6. Compute $\nabla^2 f$ in spherical coordinates for $f = r^2$.